

Jupiter

CUSTOMER CHURN ANALYSIS OF THE BANKING CLIENT 'JUPITER'

PREPARED BY:

Padmapriya Ganesh

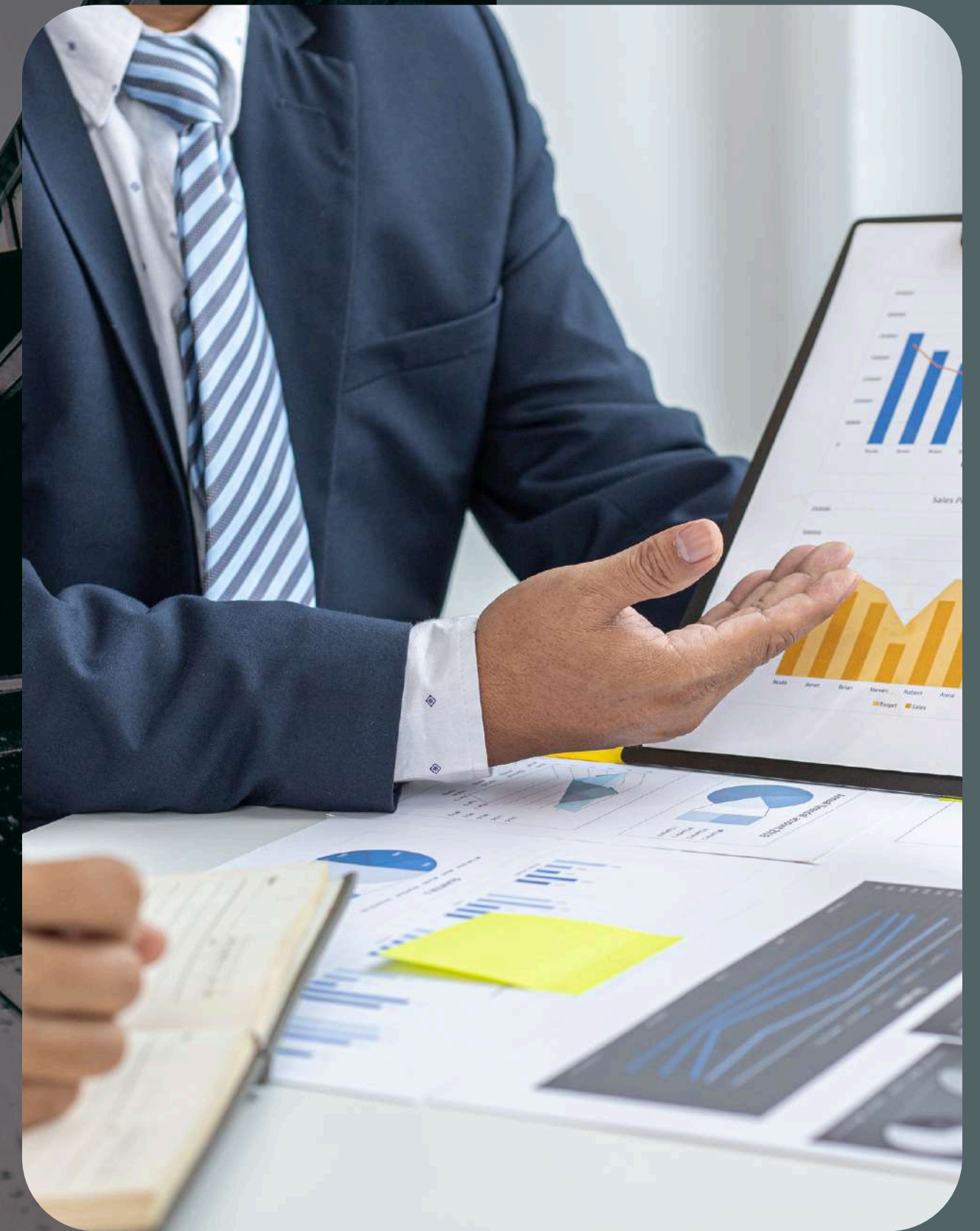
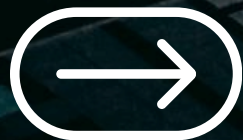
BUSINESS ANALYST

jupiter.money

Project Objective

The purpose of this analysis was to:

- Understand customer churn behavior
- Identify key factors influencing customer exit
- Determine the strongest churn indicators
- Provide business recommendations to improve customer retention



Dataset Overview



Dataset Overview

- Total customers analyzed: 10,000
- Unique customer records: 10,000
- Database Management System: PostgreSQL
- Analysis language: SQL

Customer attributes analyzed:

- Credit Score
- Geography
- Gender
- Age
- Tenure
- Balance
- Number of Products
- Active Membership
- Credit Card Ownership
- Estimated Salary
- Customer Exit Status

Target variable:

Exited (Churned / Retained Customer)

Analysis Approach

METHODOLOGY

The analysis was conducted in three stages:

1. Data Exploration

- Customer segmentation
- Churned vs retained analysis
- Single-variable analysis

2. Combined Attribute Analysis

- Geography + Age
- Geography + Products
- Geography + Membership Status
- Geography + Gender

3. Summary Statistics

- Mean
- Median
- Standard deviation
- Distribution analysis

Overall Churn Rate

Overall Customer Churn

Total customers: **10,000**

Customers exited: **2,037**

Overall churn rate:

20.37%

Approximately 1 out of every 5 customers stopped using the banking service.

Customers Churned (Exited) vs Retained

Exited,
2037, 20%



Retained,
7963, 80%

Strong Churn Indicators



Strong Influencing Factors

The following variables showed significant impact on customer churn:

- ★ Age
- ★ Geography
- ★ Number of Products
- ★ Active Membership Status

These factors created noticeable differences between retained and churned customers.

Moderate & Weak Factors



Moderate Influencing Factors

- Gender
- Balance
- Credit Score

Weak / Non-significant Factors

- Estimated Salary
- Tenure
- Credit Card Ownership

These variables showed smaller differences in churn behavior.

Key Insight #1

Focus Area 1: Improve Customer Engagement

Inactive customers showed much higher churn rates compared to active customers.

Key observation:

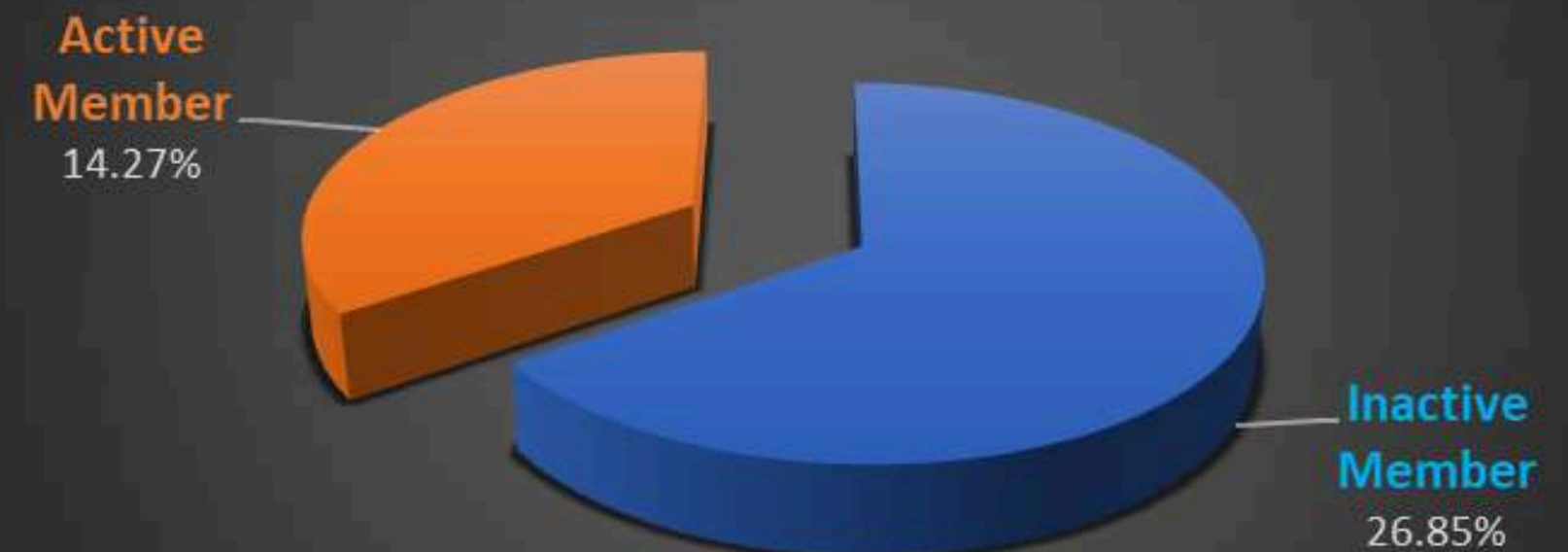
- Inactive customers → **26.85% churn**
- Active customers → **14.27% churn**

Recommended actions:

- *Loyalty programs*
- *Personalized notifications*
- *Improve customer engagement features*



Active Membership-wise customer churn %



Key Insight #2

Focus Area 2: Strengthen Retention in Germany

Germany consistently showed the highest churn across multiple analyses.

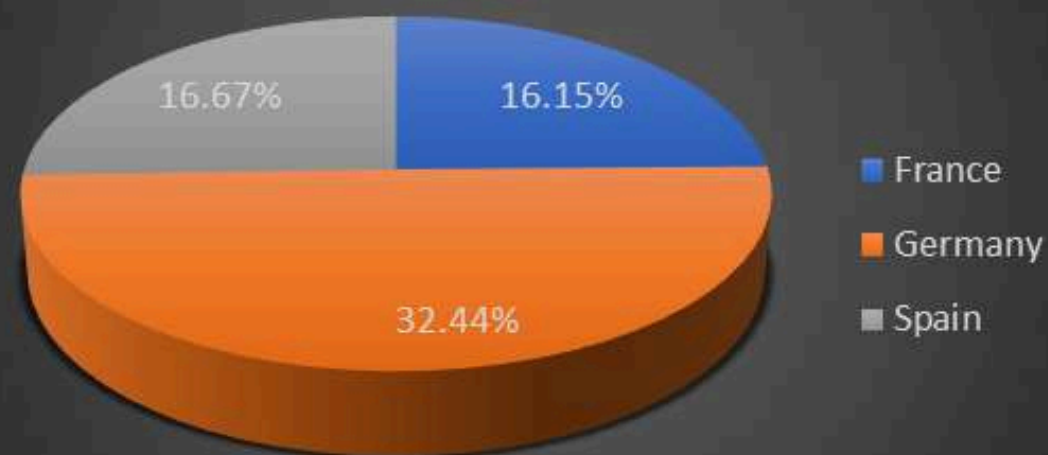
Key observation:

- Germany → **32.44% churn**
- France → **16.15% churn**
- Spain → **16.67% churn**

Recommended actions:

- *Localized campaigns*
- *Region-specific offers*
- *Improve customer support experience*

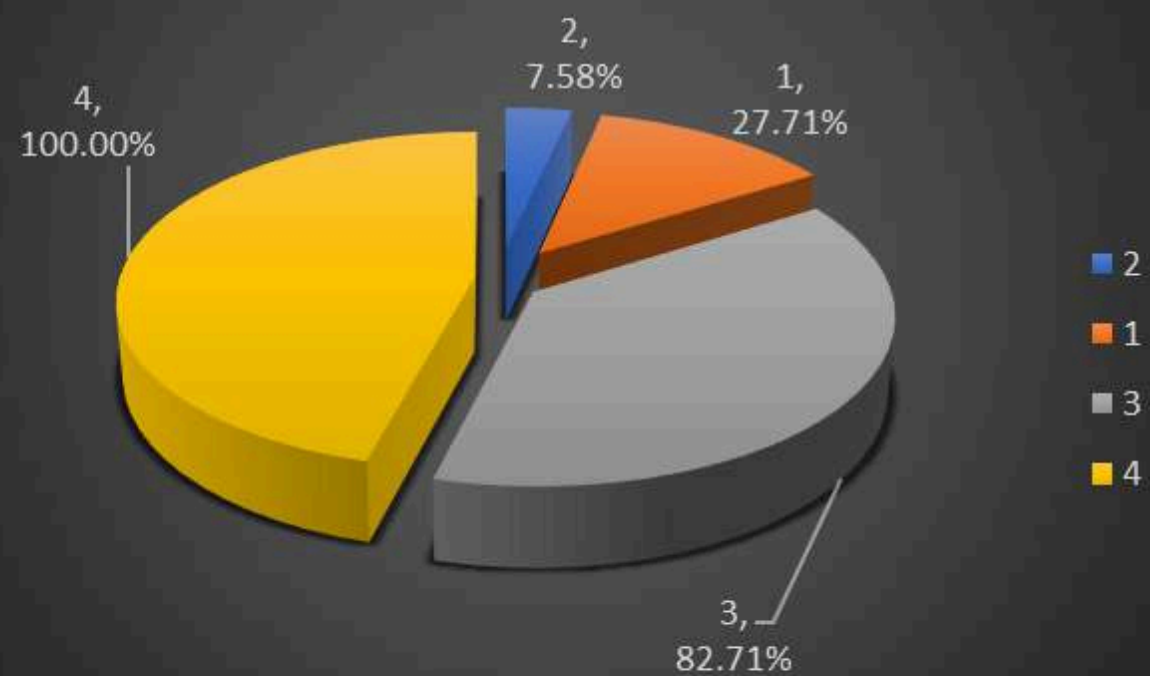
Customer Churn %



Country-wise Customer Churn

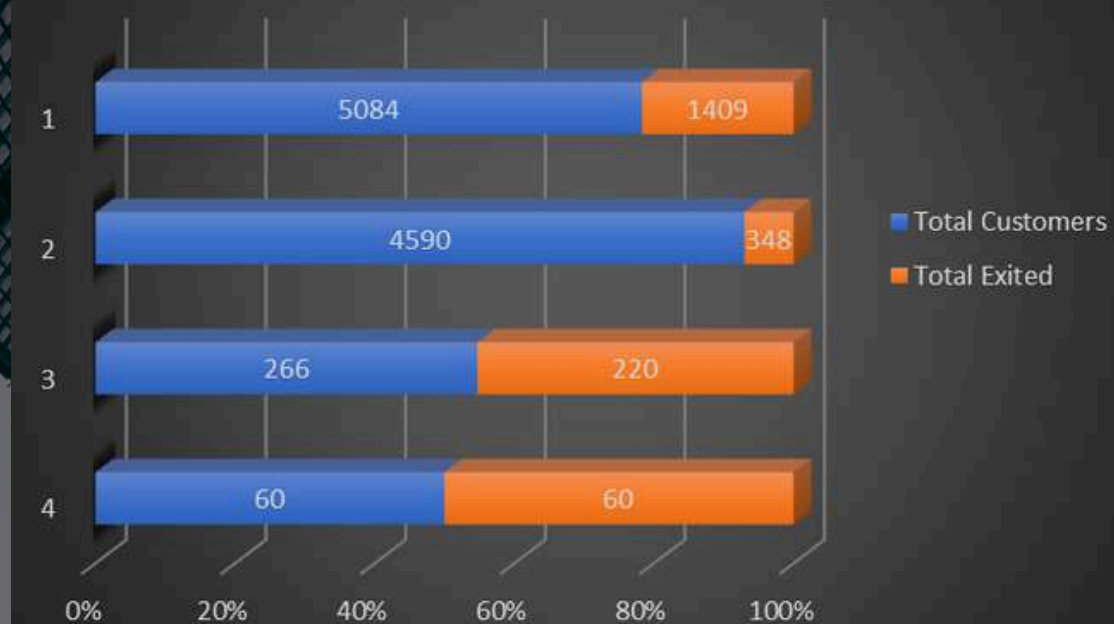


Number of products-wise customer churn %



High churn for 3-4 products may be influenced by limited customer counts.

No. of products-wise Exit vs Total



Customer product ownership strongly affected churn behavior.

Key observations:

- 1 product → **27.71% churn**
- 2 products → **7.58% churn**
- 3-4 products → very high churn rates

Recommended actions:

- *Encourage balanced product usage*
- *Create product bundles*
- *Avoid unnecessary cross-selling*

Key Insight #3

Focus Area 3:
Optimize Product Strategy



High Risk Customer Identification



Risk ranking and churn-risk scoring identified high-risk customer segments.

Common characteristics observed:

- **Germany customers**
- **Age group: 40–69 years**
- **Inactive members**
- **Product ownership: 1 or 3–4 products**

These customers may benefit from personalized retention initiatives.





Selection of Customers for Reduced Interest Rate Offers

Selection of Top 5 Customers for Personalized Retention Benefits

To reduce future customer churn, the top 5 retained customers with the highest churn risk scores were selected for reduced interest rate offers.

Selection Criteria

- Customers who have **not exited**
- High churn risk score identified using SQL window functions
- Inactive membership status
- High-risk age segment (**40–69 years**)
- High-risk product ownership pattern (**1 or 3 products**)
- Customers from high-risk regions (**Germany**)

Expected Outcome

- *Improve customer satisfaction and engagement*
- *Increase customer loyalty among high-risk customers*
- *Encourage continued usage of banking services*
- *Reduce future churn probability and improve retention rates*

Customer ID	Risk Score
15794941	9
15771728	9
15677307	8
15689294	8
15614813	8



Recommended Retention Actions

Business Recommendations

<i>Problem Identified</i>	<i>Recommended Action</i>	<i>Expected Outcome</i>
High churn among inactive customers	Introduce engagement campaigns, personalized notifications, and loyalty rewards	Increased customer activity and reduced churn
Germany showed consistently high churn rates	Launch region-specific retention campaigns and improve customer support experience	Improved retention in high-risk regions
Customers with 1 product and customers with excessive product ownership (3–4 products) showed higher churn	Encourage balanced product adoption and recommend relevant product bundles based on customer needs	Improved customer engagement and reduced churn
High-risk customers identified through churn-risk scoring	Offer personalized benefits such as reduced interest rates and targeted offers	Improved customer satisfaction and loyalty
Churn risk varies across customer segments	Use data-driven segmentation for personalized marketing and retention strategies	Better targeting and improved retention effectiveness

Jupiter

Conclusion

- The overall customer churn rate was **20.37%**
- The strongest churn drivers identified:
 - Age
 - Geography
 - Active Membership
 - Number of Products
- Data-driven analysis can help identify high-risk customers early and support targeted retention strategies.

Jupiter

THANK YOU

ANY QUESTIONS?

PREPARED BY:

Padmapriya Ganesh

BUSINESS ANALYST

jupiter.money